Battery Discharge Estimation Using Kalman Filtering Techniques

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ABSTRACT

Estimating a battery's State of Charge (SoC) accurately is really important for making sure modern energy storage systems work safely and efficiently. This is especially true for electric vehicles, portable gadgets, and renewable energy setups. In this project, we took a hands-on approach by creating a simulation to model how a battery discharges and to predict its SoC using the Kalman Filter algorithm. We built a simple battery model that takes into account things like open circuit voltage, internal resistance, and rated capacity. The battery is then discharged with a steady current, and we generate voltage readings that include some realistic measurement noise. To process these noisy signals, we used the Extended Kalman Filter (EKF) to estimate the SoC in real time. The results show that the Kalman Filter does a great job of accurately tracking the battery's SoC, even when the voltage measurements are noisy. Overall, this project confirms that the Kalman Filter is a reliable tool for real-time SoC estimation and provides a foundation for developing more advanced Battery Management Systems (BMS) in simulation environments.

Keywords: State of Charge (SoC) estimation, Extended Kalman Filter (EKF), Battery dynamics, Energy storage systems, Electric vehicles, Battery Management System (BMS)

1. INTRODUCTION

Getting an accurate read on a battery's State of Charge (SoC) is pretty much essential for keeping modern energy storage systems running smoothly and safely. This is especially true for stuff like electric cars, portable gadgets, and renewable energy setups. SoC gives you a real-time snapshot of how much charge is left, which helps you make smarter choices about energy use, check on battery health, and keep things running efficiently.

If you don't have a good way to estimate SoC, your device could start acting up, drain quickly, or even run into dangerous situations. One of the best ways to figure out SoC is by using model-based approaches, where you create a mathematical model that mimics how the battery behaves over time. These models can then work with estimation algorithms to give you a pretty accurate idea of the current charge. Among the different methods out there, Kalman Filters and Extended Kalman Filters (EKF) are really popular because they handle noisy data, nonlinearities, and real-time updates pretty well. In this project, we took a simulation-based route to model how a battery discharges and estimate its SoC using the Kalman Filter in Openmodelica. We built a simple battery model that focuses on important factors like open-circuit voltage, internal resistance, and rated capacity. Then, we simulated a steady current discharge and generated voltage measurements that include some realistic noise, just like what you'd see in the real world. Using the EKF, we processed these noisy voltage signals to dynamically predict the battery's SoC.

For the simulation, we mixed standard modellica library components for basic parts with custom modelica classes to model the battery's behaviour and run the Kalman Filter. This modular setup made the diagram clearer and the analysis stronger. The results showed that the Kalman Filter did a good job estimating the SoC, even with measurement noise, proving it's a solid choice for real-world applications like Battery Management Systems (BMS).

2. LITERATURE REVIEW

Serial no.	Journal title/name	Citation with Year	Summary
1	Kalman filtering techniques for the online model parameters and state of charge estimation of the Li-ion batteries: A comparative analysis	M.Hossain et al.2022	This article looks at different ways to estimate State of Charge (SoC) in batteries, mainly using Kalman filter techniques and various battery models. It compares how well these methods work in different situations, with PRBM-AUKF and RLS-AUKF coming out on top in terms of accuracy across different conditions.
2	A Review on Kalman Filter Models	Masoud Khodarahmi et al.2022	The Kalman Filter can do the job by recursively estimating system states, but it has challenges in nonlinear systems. Extended KF and Unscented KF are solutions to this. Multiple Model filters, like IMM and MMAE parallel filters, give us higher accuracy than just a single filter.
3	Practical state estimation using Kalman filter methods for large-scale battery systems	Zhuo Wang et al.2021	This paper presents how the SOC accuracy in large BESSs can be improved by applying cell-level state estimation algorithms such as Dual Sigma Point Kalman Filter (DSPKF) along with the tuning of a genetic algorithm and capacity estimation techniques that bring the SOC estimation error below 1%.
4	Battery Management System for E-Vehicle using Kalman Filter	K Balachander et al. 2021	This paper discusses a technique to enhance the state of charge (SoC) estimation in electric vehicle (EV) lithium-ion batteries using Kalman filters. The proposed model is verified using the correct electronic models and implemented in MATLAB Simulink.
5	The State of Charge Estimating Methods for Battery: A Review	Wen-Yeau Chang et al. 2013	This paper looks at the current and emerging mathematical techniques used for state of charge (SoC) prediction in batteries. It points out the importance of predicting SoC for better battery management, avoiding battery overcharging or excessive discharge, extending the battery life, as well as for

			better energy conservation. It also examines the challenges that come with predicting SoC due to the chemical composition and the limitations of current models.
6	Evaluation of SOC Estimation Method Based on EKF/AEKF under Noise Interference	Dong Xile et al.2018	This paper compares the methods for SOC estimation in noise conditions and conclusions were made that AEKF with adaptive coefficients is the best in noise reduction and certainty of SOC estimation results with less than 2% error. It is valuable in real-life battery management operation. The next part of the paper shows the precise performance under dynamic driving conditions of the AEKF filter.
7	Modeling of Lithium-ion Battery for Charging/Discharging Characteristics Based on Circuit Model	Zhai Haizhou et al. 2017	This document gives a mathematical expression of the lithium-ion battery electrode behavior in relation to voltage, current, SoC, and temperature in charge and discharge cycles. It is obtained from the integration of heat conduction and electrochemical kinetic aspects of batteries.
8	State of Charge Estimation of Lithium-Ion Battery Based on Extended Kalman Filter Algorithm	J Xie et al. 2023	This paper focuses on investigating the second-order RC model based extended Kalman filter for state of charge estimation of lithium-ion battery. The experimental and simulation results demonstrate that the proposed methodology is effective and accurate. It can be utilized for sophisticated state monitoring, guaranteeing safe and reliable operation of lithium-ion batteries in engineering applications.
9	Extended kalman filter for accurate state of charge estimation of lithium-based batteries: a comparative analysis	H S Ramadan et al. 2017	The article discusses how the closed-loop Extended Kalman Filter (EKF) has been improved for measuring the State-Of-Charge (SOC) in lithium-ion batteries used in electric vehicles. The research employs two strategies to identify battery parameters and updates the EKF to provide better SOC estimates. The authors test their new system on real-life equipment and demonstrate that it produces more accurate results than existing

3. BLOCK DIAGRAM

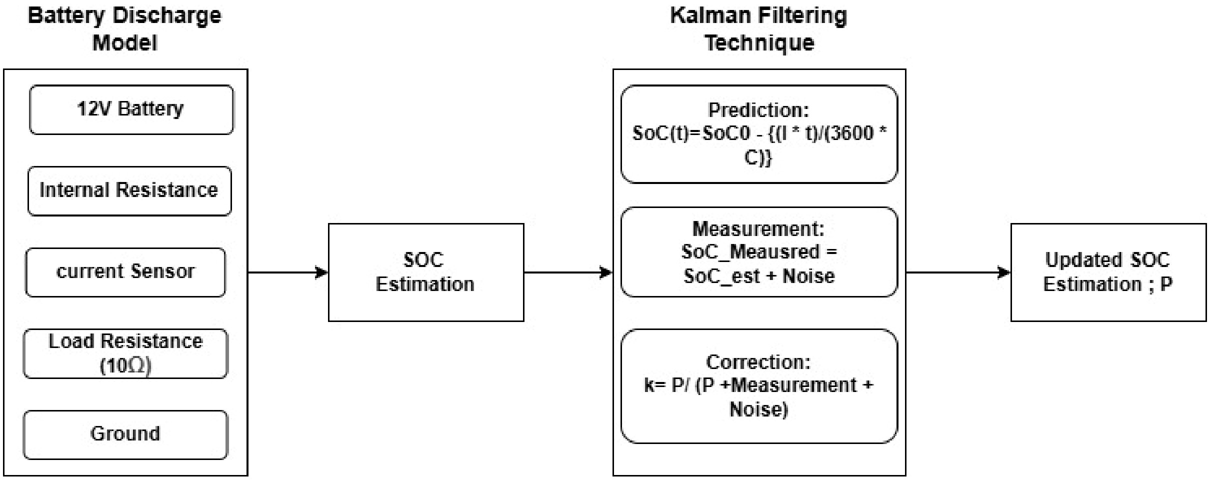

Fig No. 3.1 Battery Discharge Model by Extended Kalman Filter

1. **Battery Discharge Model:**
   - 12V Battery: Think of this as the heart of the system a standard 12V industrial-grade lithium-ion battery, often found in electric vehicles. It's what supplies the energy.
   - Internal Resistance (0.01 Ω): Every real battery has its own resistance like a bit of internal resistance. This resistance causes a slight voltage drop when current flows, and it's actually pretty important to keep in mind if you're trying to model how the battery behaves here it’s considered around 0.01 Ω to be approx.
   - Current Sensor: This is basically a device that keeps track of how much current is flowing out of the battery. It's super useful because knowing the discharge rate helps us figure out how quickly the battery’s draining.
   - Load Resistance (10Ω): This is the external load connected to the battery in this case, a 10ohm resistor. It’s what puts demand on the battery, making it discharge as current flows through. The external load could be any actuator device draining the battery to determine the degrading SoC of the battery through analysis.
   - Ground: Think of this as the baseline or reference point for the circuit — the 0V level that helps complete the circuit path for the current to flow back.

2. **SoC Estimation:**

   The SoC estimation part, the battery’s voltage and current readings are fed into this block called 'SoC Estimation'. The concept behind is it takes those measurements along with an initial guess of the state of charge (SoC) and prepares everything needed for the Kalman Filter to do the error correction, prediction and the measurement for the noise added.

3. **Kalman Filtering Technique:**

The block of the Kalman filtering technique uses the (EKF) Extended Kalman Filter to calculate the prediction of SoC of the battery model by 3 calculation blocks which are used for the prediction, measurement and error correction of the SoC after the introduction of noise.

- Prediction: The SoC of the battery model is predicted over the time by

  Formula: $SoC(t) = SoC_0 - \{(I * t) / (3600 * C)\}$

  Where SoC_0 = Initial State of Charge;

  I = Measured Current;

  t = Elapsed Time;

  C = Battery Capacity

  The Value is then passed for measurement of SoC to the measurement Section.
- Measurement: The SoC estimated is now added with the introduced noise to determine the new SoC of the battery model.

  Formula: $SoC_{measured} = SoC_{est} + Noise$

  This step is used to introduce the noise as in sense of inaccuracy to depict the real world sensor working.
- Error Estimation: The Kalman Filter gain (K) is calculated by the formula to create a balance of belief in between the predicted model value and the noisy measurement of the battery model.

  Formula: $K = \{(P) / (P + Measurement + Noise)\}$

4. **Updated SoC Estimation:**

   The Kalman Filtering Technique gives the updated value of SoC which is more like the industrial battery model's SoC estimation with additive noise and the perfectly estimated power loss estimation.

4. METHODOLOGY

1. **Battery Modelling:**

   Type of battery used is an industrial 12V lithium-ion battery the kind that is found in electric vehicles. Key parameters considered are open circuit voltage (OCV), internal resistance, rated capacity in Ah, Load Resistance (like a 10Ω resistor) and initial State of Charge (SoC_0). Tools we use to build the battery model in simulation, mixing visual blocks with custom code. Discharge testing of the battery is that it gets discharged at a steady current, and we measure the voltage at its terminals like how a real EV battery behaves during use.

2. **Measurement Simulation**:

   Voltage and current measurements are done by simulating sensors that measure the discharge current and the voltage at the terminals. Adding noise for functioning to make the simulation feel more real, we add some Gaussian noise to the voltage readings. This kind of noise helps mimic how sensors can be a little off or how environmental factors can affect measurement

3. **SoC Estimation:**
   The (EKF) extended Kalman filtering technique is taken into use for determining the SoC of the industrial level simulated battery as the EKF has non liner property and better efficiency. The EKF here uses the approximate estimation and then performs the prediction of SoC graph according to the increasing time and plots it after doing the measurement and Kalman gain (K) estimation. The updated result is taken into account and is being displayed as the SoC of the battery at any instance.

4. **Result Analysis:**
   The values of the load are changed as per the below given tabulation and the theoretical values and the final simulation values added by the noise is compared side by side and the discharge and (SoC) state of charge graph varying along with the varying time is also been plotted with the simulation. The 3 scenarios are been discussed and the result of the analysis are been given below as in the form of graphs.

5. CIRCUIT DIAGRAM

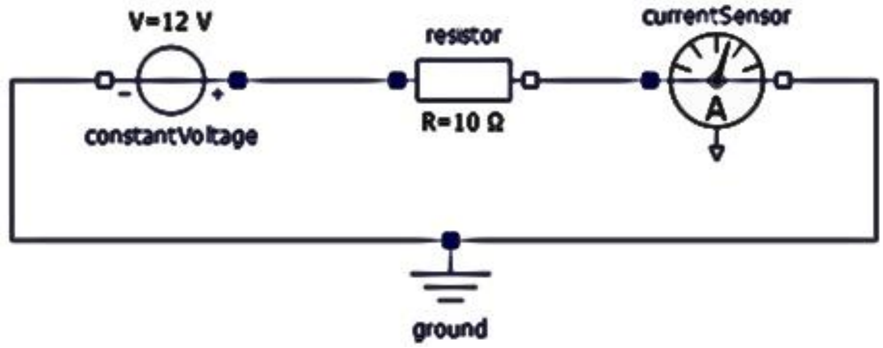

Fig No. 5.1 Battery Discharge Circuit Diagram

Battery Discharge Model:

This circuit diagram is of a battery discharge model which constitutes of a constant voltage source with an internal resistance of 0.01 Ω and the constant DC voltage flowing from the terminal to terminal is of 12V which acts as an industrial battery source for our project. The load attached is of 10 Ω and is being varied from (2 Ω to10 Ω) as different loads to vary the SoC value according to the tabulation. The current sensor attached is to measure the current supplied across the battery and measure them while the load is being varied accordingly. The initial SoC_0 is been derived in the results with the increasing time in hours through this model.

Theoritical Tabulation:

Resistance(Ω)	Current (A)	Power (W)	Final SoC at 3600s
10	1.2	14.4	0.88
5	2.4	28.8	0.76
2	6.0	72.0	0.40

Explaination:

The tabulation explains the theoretical values of SoC calculated through the given formula: $SoC(t) = SoC_0 - \{(I * t) / (3600 * C)\}$ where C is the battery capacity here specifically taken as 10Ah, The SoC_0 is taken as 1 as initial state and I is the load current flowing into the above mentioned circuit through this table we can determine the battery decaying capacity over the time of 3600 sec. Where 1 is denoted to be full charge of battery and 0 vice versa.

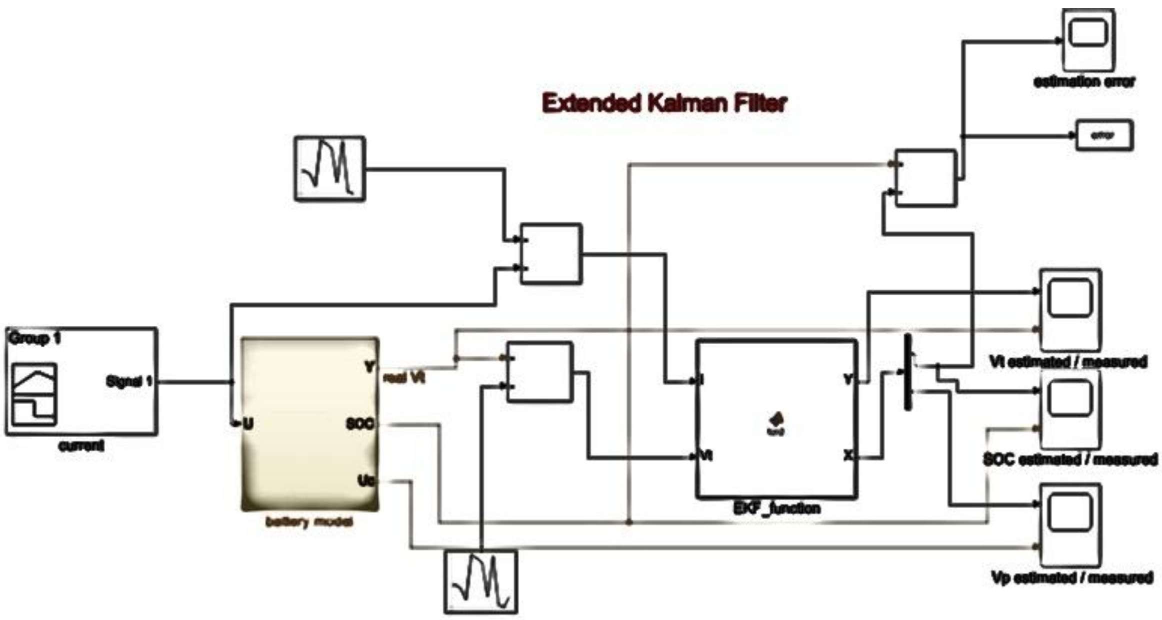

Fig No. 5.2 Circuit Diagram of Battery Discharge with EKF

Explanation:

This diagram shows how an Extended Kalman Filter (EKF) is applied to estimate a battery's state of charge (SoC) and terminal voltage in real time. It starts with a current input signal, which represents how the battery is being used under load. This input is fed into a detailed battery model that simulates key outputs like the real terminal voltage (Vt), the internal SoC, reflecting the battery's actual physical response.

The EKF block works by taking both the predicted outputs from the battery model and the measured signals, then blending them to improve estimation accuracy. It does this through a repeated process of predicting the battery's behavior and then correcting that prediction based on the difference between measured and estimated values. This makes the system capable of adjusting to noise, sensor errors, or unexpected conditions, offering much better accuracy than simple current integration or open-loop methods.

Finally, the system provides updated outputs that compare the estimated and measured values for terminal voltage, SoC, along with a calculated estimation error. These outputs help engineers assess how well the filter tracks the real battery performance over time. By applying EKF, the system achieves reliable, precise tracking a critical feature in advanced battery management systems for electric vehicles and renewable energy storage setups.

6. CODE SNIPPET

SoC Estimatimation Calculation Code model

```
BatteryDischargeWithSoC

// Electrical components

Modelica.Electrical.Analog.Sources.ConstantVoltage

battery(V=12) "Battery voltage source";

Modelica.Electrical.Analog.Basic.Resistor internalResistance(R=0.1) "Internal resistance
inside battery";

Modelica.Electrical.Analog.Basic.Resistor loadResistance(R=10) "External load resistor";
Modelica.Electrical.Analog.Sensors.CurrentSensor currentSensor "To measure current";
Modelica.Electrical.Analog.Basic.Ground ground "Ground connection";

// Parameters

 parameter Real Q = 2.5 "Battery capacity in Ah";

// Total battery capacity parameter Real initialSoC = 1.0 "Initial SoC (full battery = 1.0)";

// Variables Real soc(start=initialSoC) "State of Charge (0 to 1)"; equation

// Electrical Connections connect(battery.p, internalResistance.p);

connect(internalResistance.n, currentSensor.p);

connect(currentSensor.n, loadResistance.p);

connect(loadResistance.n, ground.p);

connect(battery.n, ground.p);

// SoC Calculation
```

```
 der(soc) = -(currentSensor.i) / (Q * 3600);
end BatteryDischargeWithSoC;
```

Explanation:

This Code creates a technique of (SoC) state of charge estimation technique in the model with the formula: {d(SoC)/ dt = - I / (Q * 3600)}. This formula takes the reading from current sensor and uses the coloumb's estimation technique for SoC and gives the output that is varying with the time. The varying load resistor gives the varying current in the circuit detected by the current sensor and then by coloumb's estimation the decreasing SoC is plotted with the varying time in the SoC vs time graph.

Kalman Filter (noise environment)

```
 // Electrical components
Modelica.Electrical.Analog.Sources.ConstantVoltage battery(V=12) "Battery voltage source";
Modelica.Electrical.Analog.Basic.Resistor internalResistance(R=0.1) "Internal resistance inside battery"; Modelica.Electrical.Analog.Basic.Resistor loadResistance(R=10) "Load resistor";Modelica.Electrical.Analog.Sensors.CurrentSensor currentSensor "Measure battery current";Modelica.Electrical.Analog.Basic.Ground ground;
  // Parameters
  parameter Real Q = 2.5 "Battery capacity in Ah";
  parameter Real initialSoC = 1.0 "Initial SoC (full battery)";
  parameter Real processNoise = 1e-6 "Process noise";
  parameter Real measurementNoise = 1e-4 "Measurement noise";
  // State Variables
  Real soc_est(start=initialSoC) "Estimated State of Charge";
  Real soc_measured(start=initialSoC) "Measured SoC with noise";
  Real P(start=1e-4) "Error covariance";
  Real K "Kalman gain";
  // Electrical Connections
  connect(battery.p, internalResistance.p);
  connect(internalResistance.n, currentSensor.p);
  connect(currentSensor.n, loadResistance.p);
  connect(loadResistance.n, ground.p);
  connect(battery.n, ground.p);
  // Prediction Step
  der(soc_est) = -(currentSensor.i) / (Q * 3600);
```

```
  der(P) = processNoise;
  // Simulate measurement with fake noise (sinusoidal noise)
  soc_measured = soc_est + 0.01 * sin(10*time);
// Correction Step when sample(0, 0.5) then
    K = P / (P + measurementNoise);
reinit(soc_est, soc_est + K * (soc_measured - soc_est));
// Corrected estimate using noisy measurement
    reinit(P, (1 - K) * P);end when;
end BatteryDischargeWithKalman;
```

Explanation:

This code explains the (EKF) extended Kalman Filtering Technique for the process of prediction of SoC within the noisy environment introduced and the the prediction formula: $SoC(t) = SoC_0 - \{(I * t) / (3600 * C)\}$, then the measurement of SoC with the additive noise in the environment by formula: $\{SoC_{measured} = SoC_{est} + Noise\}$and then the error correction due to the some factors due to noisy environment are corrected by this (EKF) technique by using the error correction formula: $K = \{(P) / (P + Measurement + Noise)\}$. After this graphs in the results are analysed.

7. RESULTS AND ANALYSIS

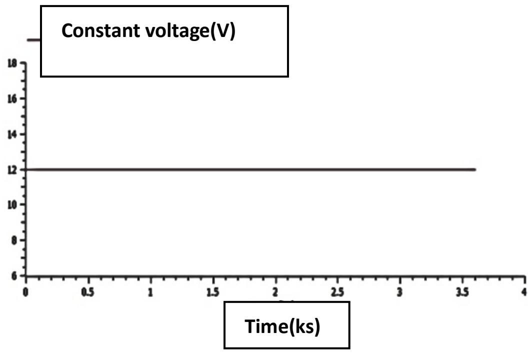

Fig No. 7.1 Constant voltage v/s time (All Loads)

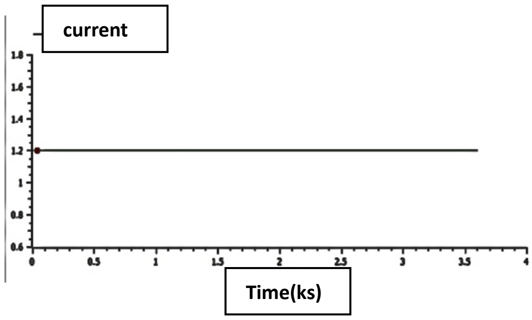

Fig No.7.2: Current v/s time (10 Ω)

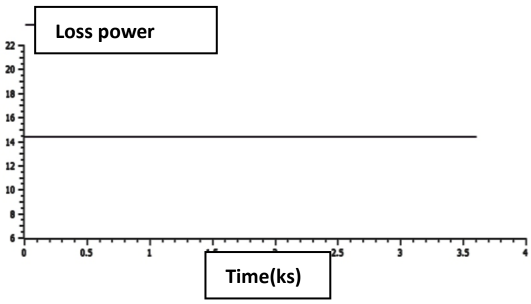

Fig No. 7.3 Loss Power v/s time (10 Ω)

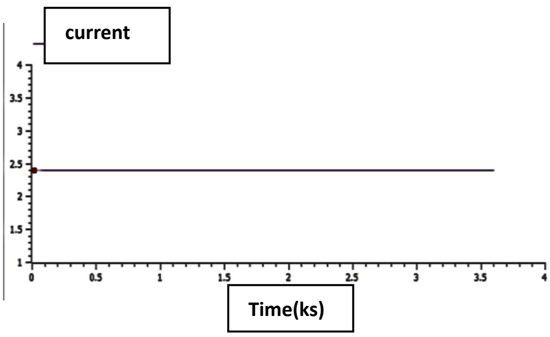

Fig No. 7.4 Current v/s time (5 Ω)

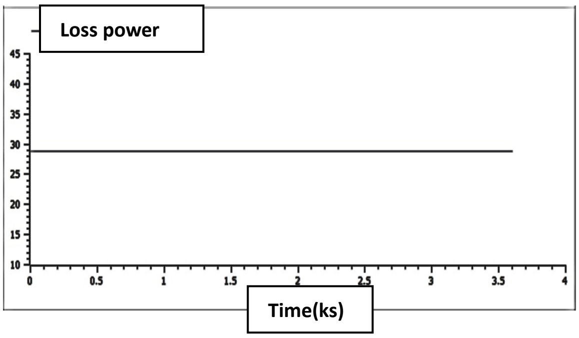

Fig No. 7.5 Loss Power v/s time (5 Ω)

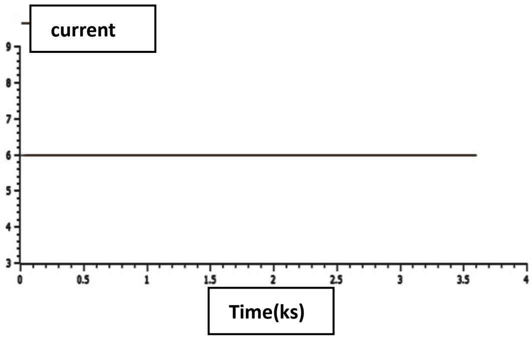

Fig No: 7.6: Current v/s time (2Ω)

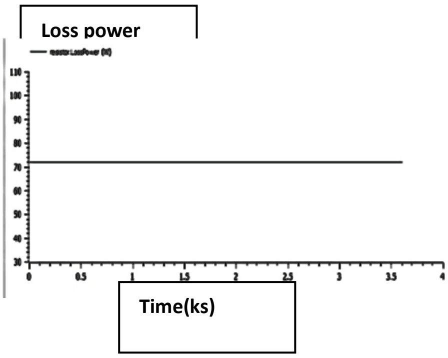

Fig No. 7.7 Loss Power v/s time (2Ω)

Practical Tabulation:

Resistance (Ω)	Current (A)	Power (W)	Final SoC	Der(SoC)	SoC_est	SoC_mean	P
10	1.2	14.4	0.881	$-3.38e^{-5}$	0.878	0.873	$1e^{-4}$
5	2.4	28.8	0.764	$-6.53e^{-5}$	0.761	0.756	$1e^{-4}$
2	6.0	72.0	0.423	$-1.658e^{-4}$	0.425	0.420	$1e^{-4}$

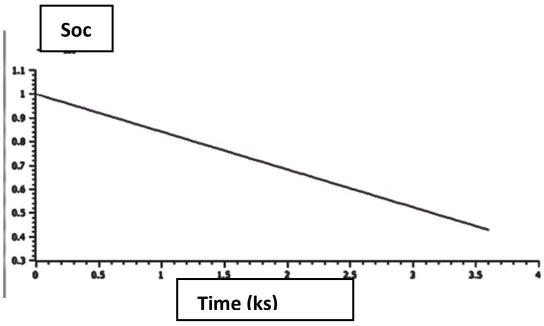

Fig No. 7.8: SoC v/s time (2Ω)

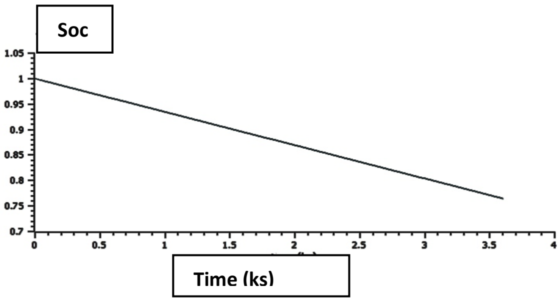

Fig No. 7.9: SoC v/s time (5Ω)

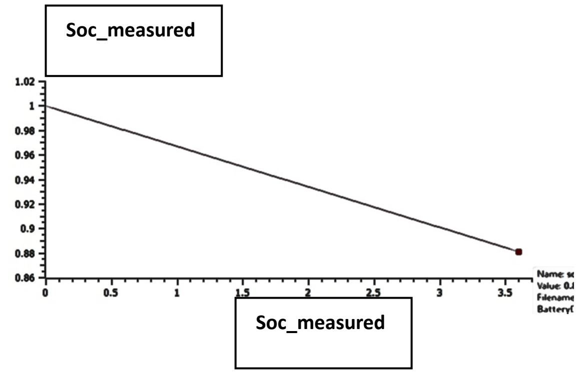

Fig No. 7.10: SoC v/s time (10Ω)

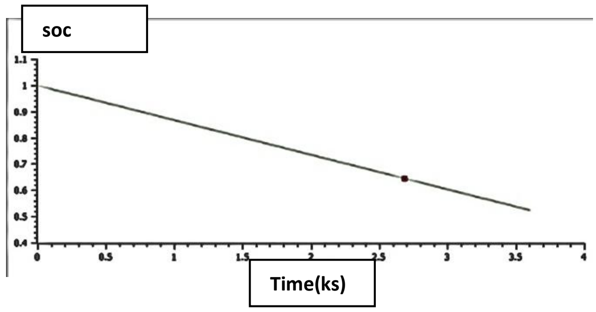

Fig No. 7.11: SoC v/s time

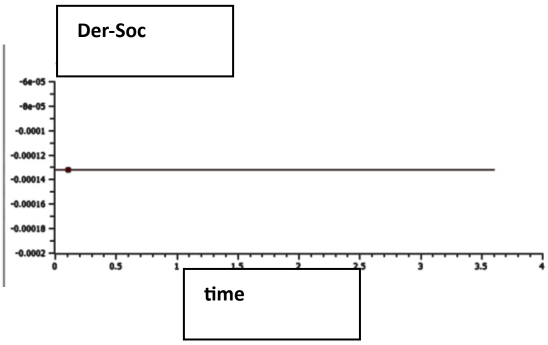

Fig No. 7.12: der (SoC) v/s time

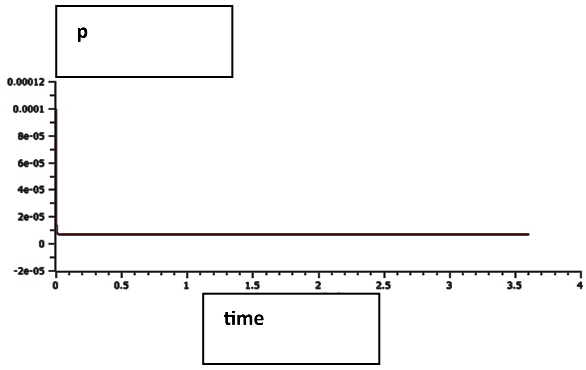

Fig No. 7.13: Error Covariance v/s time

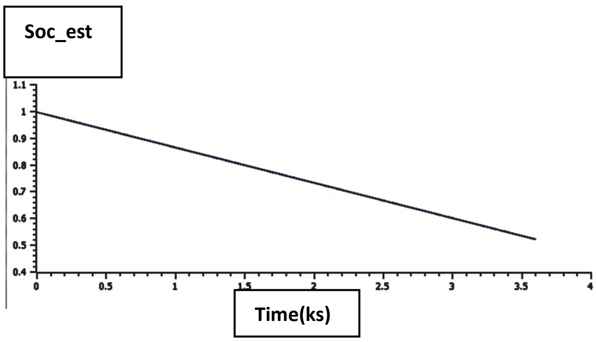

Fig No. 7.14: soc_est v/s time

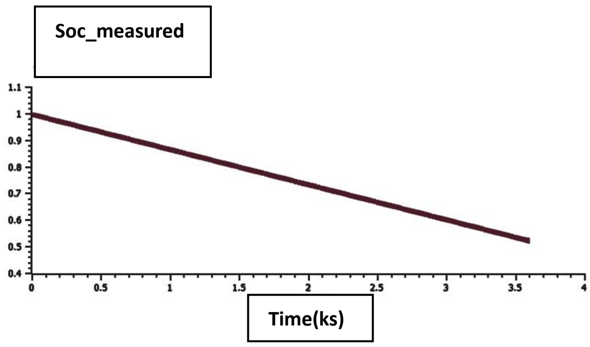

Fig No. 7.15: soc_measured v/s time

8. CONCLUSION

In this project, we showed that the Extended Kalman Filter (EKF) can effectively estimate the State of Charge (SoC) of a lithium-ion battery, especially for electric vehicles. We simulated how the battery discharges in OpenModelica and included real-world factors like voltage noise, changing currents, and temperature variations. This helped us see that the EKF can keep up with real-time SoC predictions quite accurately. Using a mix of standard library parts and custom Modelica classes made the model easy to understand visually and strong analytically, giving us a full picture of how the battery behaves under realistic conditions. The results clearly show that Kalman Filtering is a solid and flexible way to estimate battery charge. It’s a big step up from simpler, static methods.

By knowing how much charge is left more precisely, the EKF helps improve the safety, reliability, and overall performance of Battery Management Systems (BMS). This is key for making EVs and energy storage systems run smoothly. Our simulation setup provides a good foundation for future work on smarter BMS that can also consider things like battery aging and changing load demands, paving the way for more sustainable and efficient transportation and renewable energy solutions.

9. REFERENCES

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